

Math 1314 Lesson 12

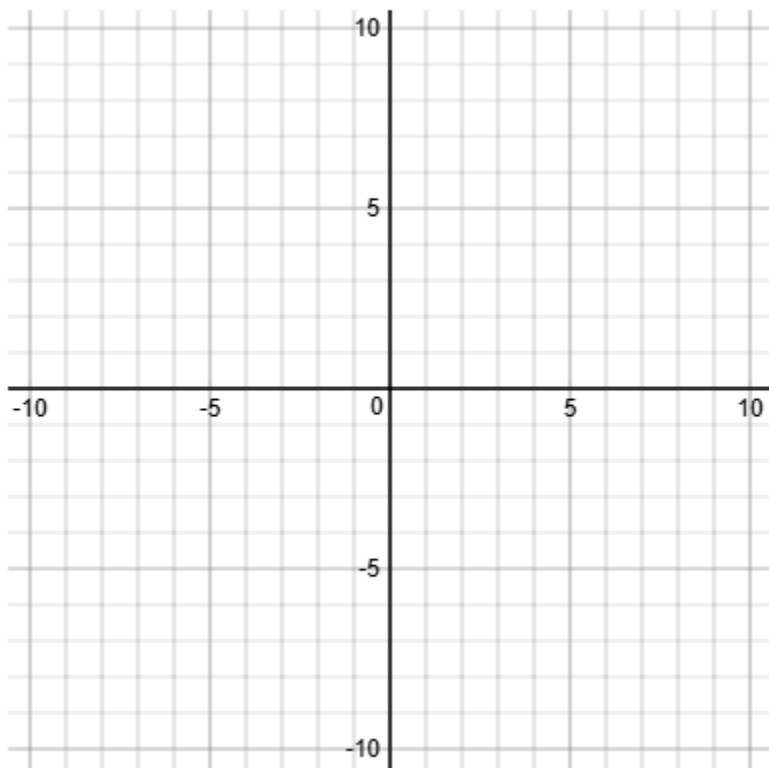
1) Evaluate the piecewise function at the given values. Graph the function.

$$f(x) = \begin{cases} 2x + 1 & \text{if } x \leq 0 \\ -3x + 2 & \text{if } x > 0 \end{cases}$$

a) $f(-3) = \underline{\hspace{2cm}} ?$

b) $f(0) = \underline{\hspace{2cm}} ?$

c) $f(4) = \underline{\hspace{2cm}} ?$



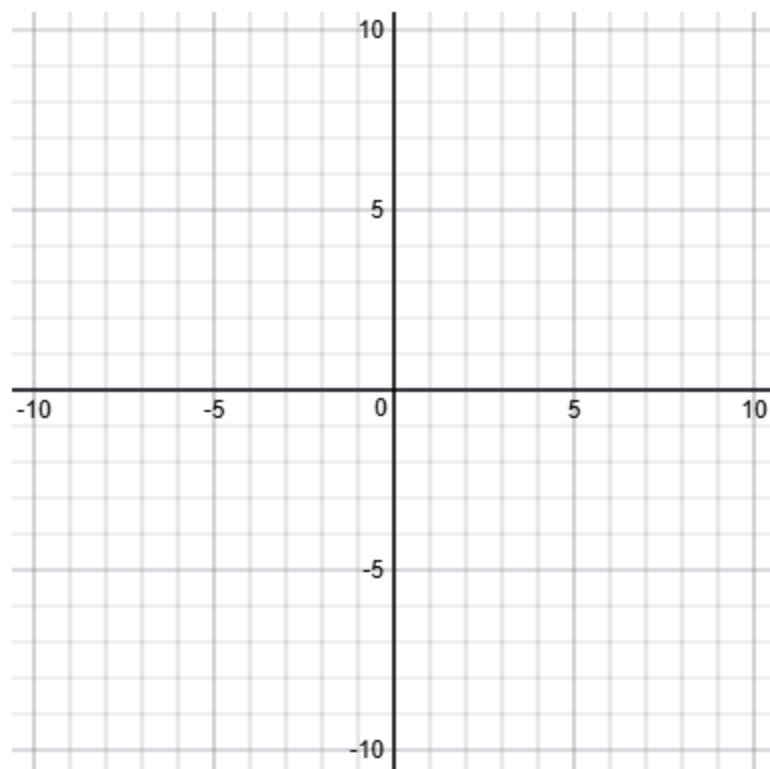
2) Evaluate the piecewise function at the given values. Graph the function.

$$f(x) = \begin{cases} 6x + 1 & \text{if } x \leq 1 \\ \frac{-x + 4}{x} & \text{if } x > 1 \end{cases}$$

a) $f(-3) = \underline{\hspace{2cm}} ?$

b) $f(0) = \underline{\hspace{2cm}} ?$

c) $f(4) = \underline{\hspace{2cm}} ?$



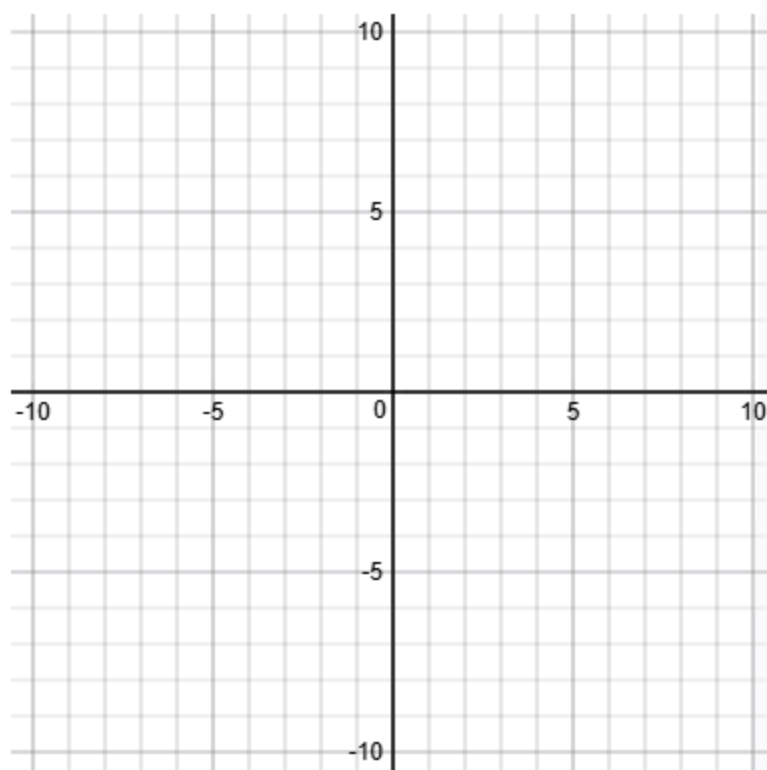
3) Evaluate the piecewise function at the given values. Graph the function.

$$f(x) = \begin{cases} 4x + 1 & \text{if } x \leq 1 \\ -5 & \text{if } 1 < x < 3 \\ 2x + 3 & \text{if } x \geq 3 \end{cases}$$

a) $f(-2) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

b) $f(1.5) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

c) $f(5) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$



4) Income Tax

Piecewise Function and 2025 Tax Brackets for Single Filers

Let x = taxable income for 2025

$$T(x) = \begin{cases} 0.10x & \text{if } 0 \leq x \leq 11,925 & (10\% \text{ Tax bracket}) \\ 1,193 + 0.12(x - 11,925) & \text{if } 11,926 \leq x \leq 48,475 & (12\% \text{ Tax bracket}) \\ 5,579 + 0.22(x - 48,475) & \text{if } 48,476 \leq x \leq 103,350 & (22\% \text{ Tax bracket}) \\ 17,651 + 0.24(x - 103,350) & \text{if } 103,351 \leq x \leq 197,300 & (24\% \text{ Tax bracket}) \\ 40,199 + 0.32(x - 197,300) & \text{if } 197,301 \leq x \leq 250,525 & (32\% \text{ Tax bracket}) \\ 57,231 + 0.35(x - 250,525) & \text{if } 250,526 \leq x \leq 626,350 & (35\% \text{ Tax bracket}) \\ 188,769.75 + 0.37(x - 626,350) & \text{if } x \geq 626,351 & (37\% \text{ Tax bracket}) \end{cases}$$

If your taxable income for the year 2025 is \$25,000, what will be your tax for the year 2025?

Note: This piecewise Tax function $T(x)$ is for Single Filers only.

For the 2025 tax year, the standard deduction is \$15,750 for single filers and married individuals filing separately, and \$31,500 for married couples filing jointly.

